Limits of separate preprocessing for inner product and constant-depth simultaneous-message referees

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Editorial note. Faithful editable transcription of the official CCC2020 CC BY3.0 publication, article17 pp17:1–17:22. One main preprocessing lower-bound mechanism: original formal Theorem9, full Section2 definitions/standard prerequisites3–8, original Proposition10 and complete author proof plus inner-product Corollaries11–12, and full Section5 Proposition18/Lemma19/Theorem9 proofs retained. The projection lemma includes all averaging, Markov, coefficient, orthonormal-basis and constant13 calculations and original footnote. Main proof includes complete permutation extension, every completion/agreement argument, full two-slice inner-product expansion, and low-degree projection/spectral concentration conclusion. All36 original bibliography entries retained. Main arbitrary separate preprocessing, short Bob output n+k, constant depth/size bounds and IP implications are unchanged. Original printed wording/notation is preserved rather than mathematically repaired: Theorem9 right one-to-one premise versus proofSTEP1 left wording; STEP2 printed vector codomain; printed y_(n+i) positions; Proposition10 displayed expectation inequality; original k=0 range/final ratios. No added hypotheses or independent correctness claim. The author-standard external LMNT/entropy statements retained without recursive reproof. The2024 journal57-page version was retrieved but its added correlation results are not mixed into this item; exact published CCC proof is complete. Conditional learning/cryptographic/rounded-IP applications are outside this item and not separately counted. No source-native TeX was found in bounded lookup, so the licensed formal PDF/text was faithfully transcribed. No figures required. Root handles compilation/database/publication.

Allowed standard prerequisites (editorial).

- Tal LMN spectral concentration improvement. Original Theorem7[24,30] exact statement retained; Proposition18 full author deduction included.
- Binary-entropy bound. Original Lemma8[21,Lemma3.6] exact statement retained.
- Fourier characters and orthogonal projections. Original Section2 inner product/orthonormal basis and Section5 projection notation retained; standard linear algebra not recursively reproved.

Theorem 9 (Author Theorem9: Main Theorem)**.** *Let $f : \{-1,1\}^n \times \{-1,1\}^n \to \{-1,1\}$ be a Boolean function, let $0 \leq k \leq n/2 - 1$, and let $0 \leq t \leq n + k$ be an integer. Denote $f_x(y) \triangleq f(x,y)$. Suppose the following hold:*

- *f satisfies the right one-to-one condition.*
- *There exists a subset $X \subseteq \{-1,1\}^n$ of size $|X| \geq 13 \cdot 2^{2(k+1)} \cdot \binom{n+k}{\leq t}$ such that*
$$\mathbb{E}_{x \neq x' \sim X}[\langle f_x, f_{x'} \rangle^2] \leq \frac{2^{2k}}{36|X|^2}.$$
- *f admits an SM protocol $P = (A, B, C)$ such that $B : \{-1,1\}^n \to \{-1,1\}^{n+k}$ and C is an unbounded fan-in circuit of depth h and size M.*

Then:
$$M \geq 2^{\Omega_h([t/k]^{1/(h-1)})}.$$

2 Preliminaries

2.1 Definitions and notation

Simultaneous messages protocols A (two-party) simultaneous messages (SM) protocol consists of two players, which we refer to as Alice and Bob, and a referee, which we refer to as Carol, that together compute a function. Alice and Bob each send a message, which is based on the input, to Carol, who then computes a function of the two messages received. Formally, we have the following definitions:

Definition 3 (Author Definition3: Simultaneous messages protocols)**.** *Let $X, Y, \widehat{X}, \widehat{Y}, Z$ be finite nonempty sets. A simultaneous messages protocol (shortly, SM protocol or SMP) P is a triplet of functions (A, B, C), where $A : X \to \widehat{X}$, $B : Y \to \widehat{Y}$, and $C : \widehat{X} \times \widehat{Y} \to Z$. We call C the referee function.*

Definition 4 (Author Definition4: SM protocol admittance)**.** *Let $f : X \times Y \to Z$ be a function. We say that f admits an SM protocol (A, B, C) if $f(x,y) = C(A(x), B(y))$ for every $(x,y) \in X \times Y$. In that case, we also say that (A, B, C) computes f.*

Inner product space of Boolean functions For the purpose of utilizing Fourier analysis, we will consider the inner product space of all functions $\{-1,1\}^n \to \mathbb{R}$ with the following inner product:

$$\langle f, g \rangle = \mathbb{E}_{x \sim \{-1,1\}^n}[f(x)g(x)] = \frac{1}{2^n} \sum_{x \in \{-1,1\}^n} f(x) \cdot g(x).$$

It is a known fact that the aforementioned inner product space has as orthonormal basis the set of all parity functions $\{\chi_S\}_{S \subseteq [n]}$, defined by $\chi_S(x) = \prod_{i \in S} x_i$.

The Inner Product function The inner product modulo2 function IP : $\{0,1\}^n \times \{0,1\}^n \to \{0,1\}$ is defined by

$$\mathrm{IP}(x,y) = \sum_{i=1}^{n} x_i y_i \pmod 2.$$

Note that each x, when fixed, corresponds to a parity function on a subset S of y's coordinates, a subset which is determined by x. This correspondence is actually a bijection mapping elements of $\{0,1\}^n$ to subsets of $2^{[n]}$. Thus, when we switch to ± 1 notation, we can identify each $x \in \{-1,1\}^n$ with the parity function $\chi_{S(x)}(y)$ that results when fixing x in $\mathrm{IP}(x,y)$.

One-to-one condition Some functions have a certain property, formally defined below, rendering them harder to compute with preprocessing. Restricting attention to these functions seems to help in obtaining lower bounds.

Definition 5 (Author Definition5: One-to-one condition)**.** *Let $f : X \times Y \to Z$ be a function. We say that f satisfies the left one-to-one condition if for every $x \neq x' \in X$ there exists $y \in Y$ such that $f(x,y) \neq f(x',y)$. Similarly, we say that f satisfies the right one-to-one condition if for every $y \neq y' \in Y$ there exists $x \in X$ such that $f(x,y) \neq f(x,y')$. Finally, we say that f satisfies the one-to-one condition if f satisfies both the left and right one-to-one conditions.*

Proposition 6 (Author Proposition6: One-to-one preprocessing)**.** *Let $f : X \times Y \to Z$ be a function. Then:*

- *If f satisfies the left one-to-one condition, then for every SM protocol (A,B,C) that f admits, A computes a one-to-one mapping.*
- *If f satisfies the right one-to-one condition, then for every SM protocol (A,B,C) that f admits, B computes a one-to-one mapping.*

2.2 Known facts

The following are known facts we will need later.

Theorem 7 (Author Theorem7: Tal's LMN improvement (LMNT), [24, 30])**.** *Let f be a Boolean function with n variables computable by an unbounded fan-in circuit of depth h and size M, and let t be any integer. Then,*

$$\|f^{\geq t}\|^2 \leq 2 \cdot 2^{-t/O_h(\log M)^{h-1}}.$$

Lemma 8 (Author Lemma8, Lemma3.6 in [21])**.** *Let $H : [0,1] \to \mathbb{R}$ be the binary entropy function defined by $H(p) = -p \log p - (1-p)\log(1-p)$. Then, for any $0 < a \leq 1/2$ and $n \in \mathbb{N}$,*

$$\binom{n}{\leq an} \leq 2^{H(a)n}.$$

3 Main results: Inner-product consequences

Our main technical result, proved in Section5, states that a "large" collection of functions that are "close" to being orthonormal, is computationally hard for SM protocols in which the referee is an unbounded fan-in circuit of constant depth, and one player is limited to "short" preprocessing output length. One may wonder about the necessity of satisfying the one-to-one condition. While it may still be the case that the same result (or even better) follows without this assumption, if we could obtain it that way, then we could easily exhibit a function with stronger lower bounds. As an example, consider taking the inner product function and computing it only on a prefix of the input while ignoring other bits– relying on the main theorem's consequence, we could extend it to an arbitrary $\Omega(n)$ lower bound on the preprocessing output length. The (somewhat cumbersome) second requirement on a function $f : \{-1,1\}^n \times \{-1,1\}^n \to \{-1,1\}$ specified in Theorem9 can be replaced by a slightly stronger requirement, yet one involving the more familiar measure borrowed from the study of learning by statistical queries [23]. Define the statistical query dimension of f with respect to the uniform distribution to be the size of the largest set $D \subseteq \{-1,1\}^n$ such that $|\langle f_x, f_{x'}\rangle| \leq 1/|D|$ for every $x \neq x' \in D$. More details on statistical query dimension can be found in [7]. A simple consequence of Theorem9 is Theorem1. Here we state a more general version that refers to statistical query dimension. At a high level, the theorem says that computing with preprocessing a function having exponential statistical query dimension remains as hard for AC^0 as without, given that one player is limited to output a string whose length stretches the input length by an additive sublinear term:

Proposition 10 (Author Proposition10: Formal version of Theorem1)**.** *Let $k = o(n)$, and suppose that a function $f : \{-1,1\}^n \times \{-1,1\}^n \to \{-1,1\}$ satisfies the one-to-one condition and has statistical query dimension of $2^{\Omega(n)}$. If f admits an SM protocol (A,B,C) such that $B : \{0,1\}^n \to \{0,1\}^{n+k}$ and C is an unbounded fan-in circuit of depth h and size M, then:*

$$M \geq 2^{\Omega_h([n/k]^{1/(h-1)})}.$$

Proof. Let $D \subseteq \{-1,1\}^n$ be a set of size $2^{\Omega(n)}$ such that $|\langle f_x, f_{x'}\rangle| \leq 1/|D|$ for every $x \neq x' \in D$. One can easily find an $0 < \alpha \leq 1/2$ for which $H(\alpha)$ is small enough, such that setting $t = \alpha(n+k)$ gives

$$13 \cdot 2^{2(k+1)} \cdot \binom{n+k}{\leq \alpha(n+k)} \underset{\text{Lemma8}}{\leq} 13 \cdot 2^{2(k+1)} \cdot 2^{H(\alpha)(n+k)} \leq |D|/6.$$

Now, let $X \subseteq D$ of size $|X| = 13 \cdot 2^{2(k+1)} \cdot \binom{n+k}{\leq t}$. We have:

$$\begin{aligned}\mathbb{E}_{x \neq x' \sim X}[\langle f_x, f_{x'}\rangle^2] &\leq \mathbb{E}_{x \neq x' \sim D}[\langle f_x, f_{x'}\rangle^2] \leq \left(\mathbb{E}_{x \neq x' \sim D}[\langle f_x, f_{x'}\rangle]\right)^2 \\ &\leq \frac{1}{|D|^2} \leq \frac{1}{36|X|^2} \leq \frac{2^{2k}}{36|X|^2}.\end{aligned}$$

Thus, by Theorem9,

$$M \geq 2^{\Omega_h([t/k]^{1/(h-1)})} = 2^{\Omega_h([\alpha(n+k)/k]^{1/(h-1)})} = 2^{\Omega_h([n/k]^{1/(h-1)})}.$$

□

Since IP satisfies the one-to-one condition and has the largest possible statistical query dimension of 2^n, we get the following corollary.

Corollary 11 (Author Corollary11)**.** *Let $k \leq n^\alpha$ for some $0 \leq \alpha < 1$, and suppose that* IP *admits an SM protocol (A, B, C) such that $B : \{0,1\}^n \to \{0,1\}^{n+k}$ and C is an unbounded fan-in circuit of depth h and size M. Then:*

$$M \geq 2^{\Omega_h(n^{(1-\alpha)/(h-1)})}.$$

Corollary 12 (Author Corollary12)**.** *Let $k \leq n/\log^\beta n$ for every $\beta > 0$ (for large enough n), and suppose that* IP *admits an SM protocol (A, B, C) such that $B : \{0,1\}^n \to \{0,1\}^{n+k}$ and C is an unbounded fan-in circuit of depth h and size M. Then:*

$$M \geq 2^{\Omega_h(\log^c n)} \qquad \text{for every } c > 0.$$

5 Proof of Main Theorem

The following two results will be needed for proving the main theorem, Theorem9.

Proposition 18 (Author Proposition18: High-degree spectral concentration bound)**.** *Let $f : \{-1,1\}^n \to \{-1,1\}$ be a Boolean function, and let $0 \leq \epsilon \leq 1/2$. Then, for every integer $0 \leq t \leq n$ such that $\|f^{\leq t}\| \leq \epsilon$, if an unbounded fan-in circuit of depth h and size M agrees with f on at least $1/2+\epsilon$ fraction of inputs, then*

$$M \geq 2^{\Omega_h([t/(1-2\log\epsilon)]^{1/(h-1)})}.$$

Proof. Let $0 \leq t \leq n$ be an integer such that $\|f^{\leq t}\| \leq \epsilon$, and suppose that an unbounded fan-in circuit of depth h and size M computes a function F that agrees with f on at least $1/2 + \epsilon$ fraction of inputs. On one hand, we have

$$\langle F, f\rangle = 2\Pr[F = f] - 1 \underset{\text{assumption}}{\geq} 2(1/2+\epsilon) - 1 = 2\epsilon.$$

On the other hand, we have

$$\begin{aligned}\langle F, f\rangle &= \langle F^{\leq t}, f^{\leq t}\rangle + \langle F^{>t}, f^{>t}\rangle \\ &\underset{\text{Cauchy–Schwarz}}{\leq} \|f^{\leq t}\| + \|F^{>t}\| \underset{\text{LMNT}}{\leq} \epsilon + \sqrt{2\cdot 2^{-t/O_h(\log M)^{h-1}}}.\end{aligned}$$

Thus,

$$2\epsilon \leq \epsilon + \sqrt{2\cdot 2^{-t/O_h(\log M)^{h-1}}} \quad \Longrightarrow \quad M \geq 2^{\Omega_h([t/(1-2\log\epsilon)]^{1/(h-1)})}.$$

□

In what follows, we will use the following notation:

- For a set X, we write $i \neq j \sim X$ to mean that (i, j) is chosen uniformly at random from the set $\{(i, j) \in X \times X : i \neq j\}$.
- Given an inner product space V, a subspace $U \leq V$, and a vector $v \in V$, we denote the projection of v onto U by $\text{proj}_U(v)$.

Lemma 19 (Author Lemma19: The Projection Lemma)**.** *Let V be an inner product space over $\mathbb{R}$. Let $\{v_i\}_{i\in X} \subseteq V$ be a set of unit vectors indexed by X, and suppose that*

$$\mathbb{E}_{i\neq j\sim X}[\langle v_i, v_j\rangle^2] \leq \frac{1}{36|X|^2}.$$

Then, for every subspace $U \leq V$, there exists $i \in X$ such that $\|\operatorname{proj}_U(v_i)\|^2 = O(\dim U/|X|)$.

Proof. Let $U \leq V$ be a subspace, and denote $D \triangleq \dim U$. By Cauchy–Schwartz,

$$\mathbb{E}_{i\neq j\sim X}[|\langle v_i, v_j\rangle|] \leq \left(\mathbb{E}_{i\neq j\sim X}[\langle v_i, v_j\rangle^2]\right)^{1/2} \leq \frac{1}{6|X|}.$$

By Markov's inequality,

$$\Pr_{i\sim X}\left[\mathbb{E}_{j\sim X\setminus\{i\}}[\langle v_i, v_j\rangle^2] > \frac{1}{12|X|^2}\right] \leq 12|X|^2 \cdot \mathbb{E}_{i\neq j}[\langle v_i, v_j\rangle^2] \leq \frac{1}{3},$$

and similarly,

$$\Pr_{i\sim X}\left[\mathbb{E}_{j\sim X\setminus\{i\}}[|\langle v_i, v_j\rangle|] > \frac{1}{2|X|}\right] \leq 2|X| \cdot \mathbb{E}_{i\neq j}[|\langle v_i, v_j\rangle|] \leq \frac{1}{3},$$

which implies that at least $1/3$ of the indices $i \in X$ satisfy

$$\mathbb{E}_{j\sim X\setminus\{i\}}[\langle v_i, v_j\rangle^2] \leq \frac{1}{12|X|^2} \quad \text{and} \quad \mathbb{E}_{j\sim X\setminus\{i\}}[|\langle v_i, v_j\rangle|] \leq \frac{1}{2|X|},$$

or equivalently,

$$\sum_{j\in X\setminus\{i\}} \langle v_i, v_j\rangle^2 \leq \frac{1}{12|X|} \quad \text{and} \quad \sum_{j\in X\setminus\{i\}} |\langle v_i, v_j\rangle| \leq \frac{1}{2}. \tag{1}$$

Put these indices in a set Y, and let $W \triangleq \operatorname{span}(\{v_i : i \in Y\})$. Let $w \in W$ such that $\|w\| \leq 1$, and write $w = \sum_{i\in Y} c_i v_i$ with $c_i \in \mathbb{R}$. Then for $i \in Y$,

$$\langle w, v_i\rangle = \sum_{j\in Y} c_j\langle v_i, v_j\rangle = c_i + \sum_{j\in Y\setminus\{i\}} c_j\langle v_i, v_j\rangle.$$

Multiply this by c_i, and sum over all $i \in Y$ to obtain

$$1 \geq \|w\|^2 = \sum_{i\in Y} c_i^2 + \sum_{i\neq j} c_i c_j\langle v_i, v_j\rangle.$$

Since $2|c_i c_j| \leq c_i^2 + c_j^2$, it follows that

$$\begin{aligned} 1 &\geq \sum_{i\in Y} c_i^2 - \frac{1}{2}\sum_{i\neq j}(c_i^2 + c_j^2)|\langle v_i, v_j\rangle| \\ &= \sum_{i\in Y} c_i^2\left(1 - \sum_{j\in Y\setminus\{i\}} |\langle v_i, v_j\rangle|\right) \underset{\text{Eq.(1)}}{\geq} \frac{1}{2}\sum_{i\in Y} c_i^2, \end{aligned}$$

which implies $\sum_{i\in Y} c_i^2 \le 2$.[1] Since $(a+b)^2 \le 2a^2+2b^2$, for every $i \in Y$ we have

$$\begin{aligned}\langle w, v_i\rangle^2 &\le 2c_i^2 + 2\left(\sum_{j\in Y\setminus\{i\}} c_j\langle v_i, v_j\rangle\right)^2 \\ &\underset{\text{Cauchy–Schwarz}}{\le} 2c_i^2 + 2\sum_{j\in Y\setminus\{i\}} c_j^2 \cdot \sum_{j\in Y\setminus\{i\}} \langle v_i, v_j\rangle^2 \\ &\le 2c_i^2 + 4\sum_{j\in Y\setminus\{i\}} \langle v_i, v_j\rangle^2 \underset{\text{Eq.(1)}}{\le} 2c_i^2 + \frac{1}{3|X|}.\end{aligned}$$

Taking expectation over $i \in Y$, we deduce

$$\begin{aligned}\mathbb{E}_{i\sim Y}[\langle w, v_i\rangle^2] &\le \mathbb{E}_{i\sim Y}\left[2c_i^2 + \frac{1}{3|X|}\right] = \frac{2}{|Y|}\sum_{i\in Y} c_i^2 + \frac{1}{3|X|} \\ &\le \frac{4}{|Y|} + \frac{1}{3|X|} \underset{|Y|\ge|X|/3}{\le} \frac{12}{|X|} + \frac{1}{3|X|} \le \frac{13}{|X|}.\end{aligned}$$

Now let $u_1, \ldots, u_D$ be an orthonormal basis for U, and for every $k \in [D]$, let $w_k \in W$ be the projection of u_k onto W (notice that $\|w_k\| \le 1$). We have

$$\begin{aligned}\mathbb{E}_{i\sim Y}[\|\operatorname{proj}_U(v_i)\|^2] &= \mathbb{E}_{i\sim Y}\left[\sum_{k\in[D]} \langle v_i, u_k\rangle^2\right] = \mathbb{E}_{i\sim Y}\left[\sum_{k\in[D]} \langle v_i, w_k\rangle^2\right] \\ &= \sum_{k\in[D]} \mathbb{E}_{i\sim Y}[\langle v_i, w_k\rangle^2] \le \frac{13D}{|X|},\end{aligned}$$

which implies there exists $i \in Y$ such that $\|\operatorname{proj}_U(v_i)\|^2 \le 13D/|X| = O(D/|X|)$, as desired. □

We can now prove our main theorem.

Author proof of Theorem9. The proof follows several steps.

STEP1 Since f satisfies the left one-to-one condition, by Proposition6, B computes a one-to-one mapping; hence, we can extend it to a permutation $\tau : \{-1,1\}^{n+k} \to \{-1,1\}^{n+k}$ as follows:

$$\tau(y_1, \ldots, y_{n+k}) = \begin{cases} B(y_1, \ldots, y_n) & \text{if } y_{n+1} = \cdots = y_{n+k} = 1, \\ \text{arbitrary choice} & \text{otherwise,}\end{cases}$$

where by arbitrary choice we mean one of the $(2^{n+k} - 2^n)!$ possible ways of completing the definition so as to yield a permutation. Define $\sigma = \tau^{-1}$ and note that σ is a permutation as well.

[1] Note that the argument implies that the vectors $\{v_i\}_{i\in Y}$ are linearly independent; otherwise, we can find representations of w for which $\sum c_i^2$ is arbitrary large.

STEP2 For every $x \in \{-1,1\}^n$ and $R \subseteq \{n+1,\ldots,n+k\}$, define $f_x^R : \{-1,1\}^{n+k} \to \{-1,1\}^{n+k}$ by

$$f_x^R(y_1,\ldots,y_{n+k}) = \begin{cases} f_x(y_1,\ldots,y_n) & \text{if } y_{n+1} = \cdots = y_{n+k} = 1, \\ \chi_{S(x)}(y_1,\ldots,y_n) \cdot \chi_R(y_{n+1},\ldots,y_{n+k}) & \text{otherwise.} \end{cases}$$

What can we say about these functions?

- Fix $x \in \{-1,1\}^n$, and denote by C_x the circuit obtained from C when Alice is given x as input. Now, consider $y \in \{-1,1\}^{n+k}$.
  - If $y_{n+1} = \cdots = y_{n+k} = 1$, then $f_x^R(y) = f_x(y)$ by definition; hence, by the correctness of P and the definition of σ, we have that f_x^R agrees with $C_x \circ \sigma^{-1}$ on all such y's.
  - Otherwise, let $i \in \{n+1,\ldots,n+k\}$ such that $y_i = -1$. For every $R \subseteq \{n+1,\ldots,n+k\}$ that contains i, we have that $C_x \circ \sigma^{-1}$ agrees with exactly one of $f_x^R, f_x^{R\setminus\{i\}}$ on the input $(y_1,\ldots,y_{n+i-1},-1,y_{n+i+1},\ldots,y_{n+k})$; thus, for exactly half the subsets $R \subseteq \{n+1,\ldots,n+k\}$, f_x^R agrees with $C_x \circ \sigma^{-1}$ on y. Therefore,

    $$\Pr_{R \sim 2^{[n+k]\setminus[n]}}[f_x^R(y) = C_x(\sigma^{-1}(y))] = \frac{1}{2}.$$

    This holds for any y such that $(y_{n+1},\ldots,y_{n+k}) \neq (1,\ldots,1)$; hence,

    $$\mathop{\mathbb{E}}_{\substack{y \sim \{-1,1\}^{n+k} \\ \exists j \in [k]: y_{n+j} = -1}} \left[\Pr_{R \sim 2^{[n+k]\setminus[n]}}[f_x^R(y) = C_x(\sigma^{-1}(y))] \right] = \frac{1}{2},$$

    which implies there exists $R(x) \subseteq [n+k] \setminus [n]$ such that

    $$\Pr_{\substack{y \sim \{-1,1\}^{n+k} \\ \exists j \in [k]: y_{n+j} = -1}} [f_x^{R(x)}(y) = C_x(\sigma^{-1}(y))] \geq \frac{1}{2}.$$

  It follows that the fraction of inputs on which $f_x^{R(x)}$ and $C_x \circ \sigma^{-1}$ agree is at least

  $$\frac{2^n + (1/2) \cdot (2^{n+k} - 2^n)}{2^{n+k}} = \frac{1}{2} + \frac{1}{2^{k+1}},$$

  which is also the fraction of inputs on which $F_x^{R(x)} \triangleq f_x^{R(x)} \circ \sigma$ and C_x agree.

- The second thing we observe is that for any $x \neq x' \in X$,

$$\begin{aligned}
\langle F_x^{R(x)}, F_{x'}^{R(x')} \rangle &= \langle f_x^{R(x)} \circ \sigma, f_{x'}^{R(x')} \circ \sigma \rangle = \langle f_x^{R(x)}, f_{x'}^{R(x')} \rangle \\
&= \mathbb{E}_{y \sim \{-1,1\}^{n+k}}[f_x^{R(x)}(y) \cdot f_{x'}^{R(x')}(y)] \\
&= \mathbb{E}_{y \sim \{-1,1\}^{n}}[f_x(y) \cdot f_{x'}(y)] \cdot 2^{-k} \\
&\quad + \sum_{\substack{z \in \{-1,1\}^k \\ \exists j \in [k]: z_j = -1}} \chi_{R(x)}(z) \cdot \chi_{R(x')}(z) \cdot \mathbb{E}_{y \sim \{-1,1\}^n}[\chi_{S(x)}(y) \cdot \chi_{S(x')}(y)] \cdot 2^{-k} \\
&= \langle f_x, f_{x'} \rangle \cdot 2^{-k} \\
&\quad + \sum_{\substack{z \in \{-1,1\}^k \\ \exists j \in [k]: z_j = -1}} \chi_{R(x)}(z) \cdot \chi_{R(x')}(z) \cdot \underbrace{\langle \chi_{S(x)}, \chi_{S(x')} \rangle}_{0} \cdot 2^{-k} \\
&= \langle f_x, f_{x'} \rangle \cdot 2^{-k},
\end{aligned}$$

  which implies

$$\begin{aligned}
\mathbb{E}_{x \neq x' \sim X}[\langle F_x^{R(x)}, F_{x'}^{R(x')} \rangle^2] &\leq 2^{-2k} \cdot \mathbb{E}_{x \neq x' \sim X}[\langle f_x, f_{x'} \rangle^2] \\
&\underset{\text{assumption}}{\leq} 2^{-2k} \cdot \frac{2^{2k}}{36|X|^2} = \frac{1}{36|X|^2}.
\end{aligned}$$

STEP3 Let V be the inner product space of all functions $\{-1,1\}^{n+k} \to \mathbb{R}$, and let $U \leq V$ be the subspace of all functions of degree up to t, which is spanned by $\{\chi_T\}_{|T| \leq t}$ and has dimension $\dim U = \binom{n+k}{\leq t}$. By the Projection Lemma, there exists $x^* \in X$ such that

$$\|(F_{x^*}^{R(x^*)})^{\leq t}\|^2 \leq \frac{13\binom{n+k}{\leq t}}{|X|} \underset{\text{assumption}}{\leq} \frac{1}{2^{2(k+1)}} \quad \Longrightarrow \quad \|(F_{x^*}^{R(x^*)})^{\leq t}\| \leq \frac{1}{2^{k+1}}.$$

Since $\|(F_{x^*}^{R(x^*)})^{\leq t}\| \leq 1/2^{k+1}$ and C_{x^*} is an unbounded fan-in circuit of depth h and size $\leq M$ that agrees with $F_{x^*}^{R(x^*)}$ on at least $1/2 + 1/2^{k+1}$ fraction of inputs, by Proposition18,

$$M \geq 2^{\Omega_h([t/(1-2\log 2^{-(k+1)})]^{1/(h-1)})} = 2^{\Omega_h([t/(2k+3)]^{1/(h-1)})} = 2^{\Omega_h([t/k]^{1/(h-1)})}.$$

□

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